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(74) Ambiguous Grammar

A Grammar is said to be Ambiguous if there exists two or more derivation tree for a string w (that means two or more left derivation trees)

Example: $G = (\{S\}, \{a+b, +, *\}, P, S\}$

$P: S \rightarrow S+S \mid S*S \mid a \mid b$

→ want $a+a*b$

① $S \rightarrow \boxed{S+S}$

$\rightarrow a+S \quad (S \rightarrow a)$

$\rightarrow a+S*S \quad (S \rightarrow S*S)$

$\rightarrow a+a*S \quad (S \rightarrow a)$

$\rightarrow a+a*b \quad (S \rightarrow b)$

② $S \rightarrow \boxed{S*S}$

$\rightarrow S+S*S$ Thus, ambiguous

$\rightarrow a+S*S$

$\rightarrow a+a*S$

$\rightarrow a+a*b$